Lecture 14 (10/16)

Simpson's paradox and regression with many predictors

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- Examples (violent crime and baseball)

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Today:

1) Assignment 2
2) Simpson's paradox: batting averages
3) Omitted variables in autoregressive processes

Assignment 2: Gaza casualty counts

Wyner argues the Gaza Health Ministry's daily casualty counts are fabricated; Pachter's blog post responds to his first claim.

1) In the tablet article, what are some of the statistical claims and are they justified? Is there evidence that "The total civilian casualty count is likely to be extremely overstated"?

Assignment 2: Gaza casualty counts

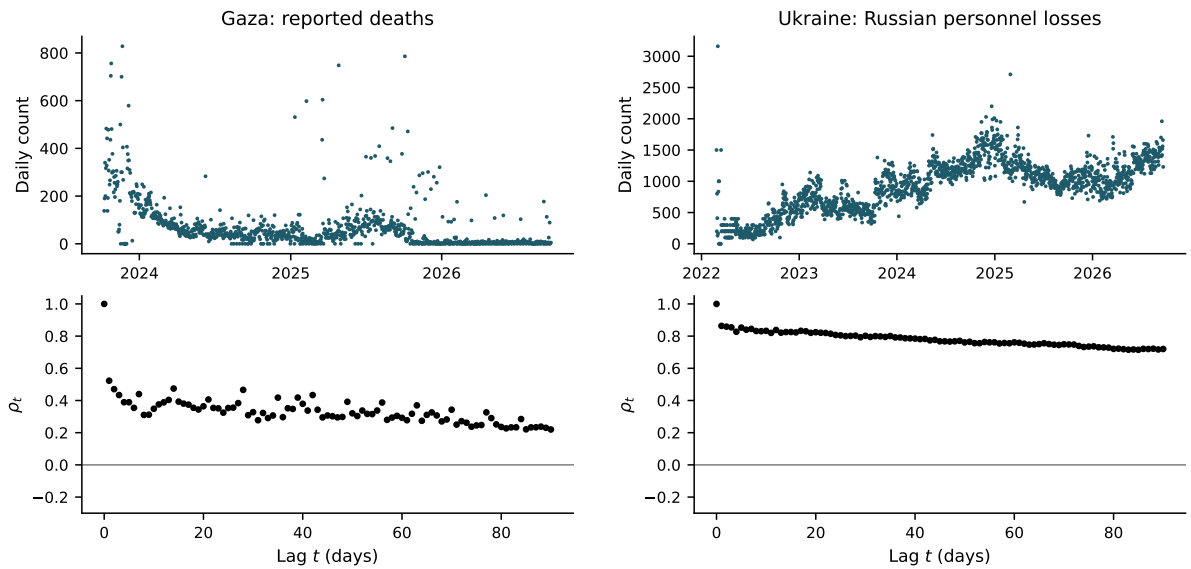
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1) In the tablet article, what are some of the statistical claims and are they justified? Is there evidence that "The total civilian casualty count is likely to be extremely overstated"?
2) Does the "Garden of forking paths" apply to this analysis?
3) Is the following plot of the lagged correlations of daily deaths in the Ukraine and Gaza dataset to be expected? Why or why not?



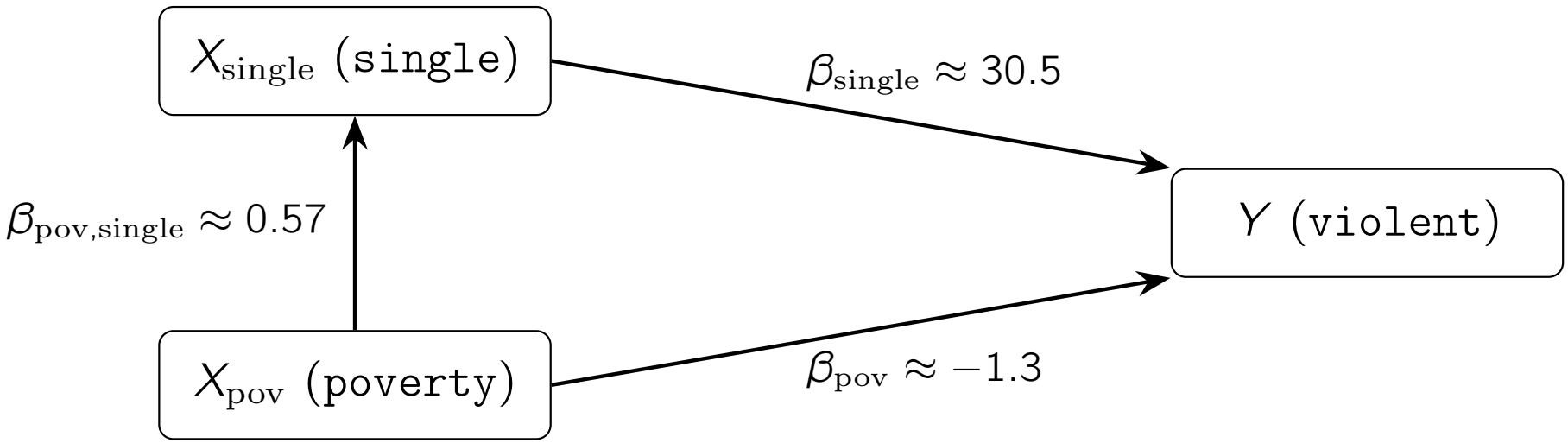
Crime data review

Recall from last class:

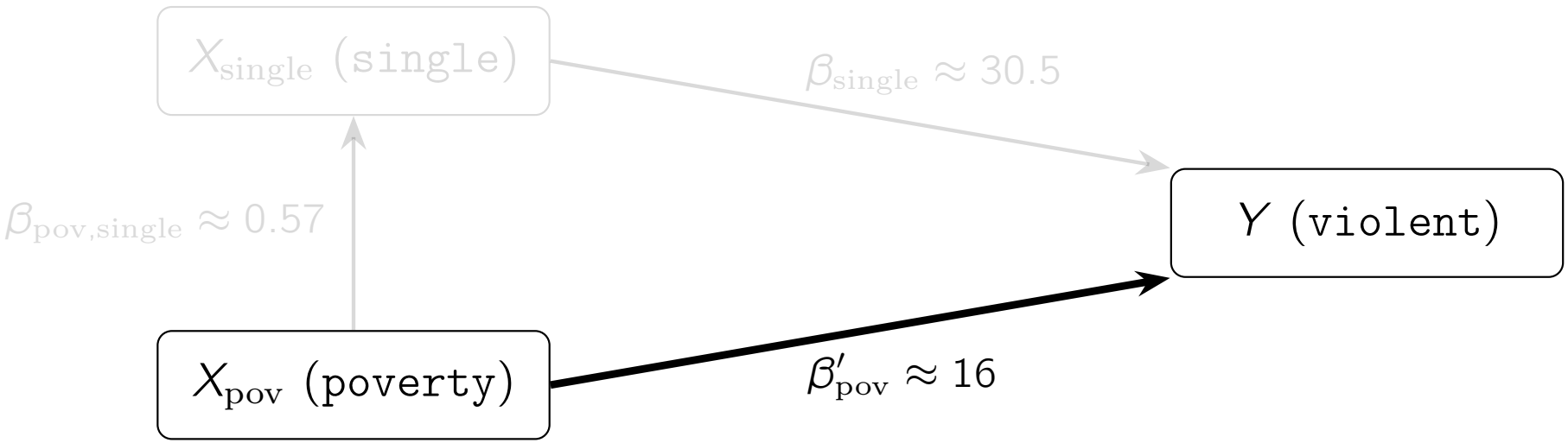
- Y (`violent`): violent crimes per 100,000 people;
- X_{pov} (`poverty`): % of people below the poverty line;
- X_{single} (`single`): % of family households headed by a single parent.

```
Dep. Variable:                violent   R-squared:                       0.093
                 coef    std err          t      P>|t|      [0.025      0.975]
const        172.9711    101.400      1.706      0.095     -30.906     376.849
poverty       15.9665      7.193      2.220      0.031       1.503      30.430
Dep. Variable:                violent   R-squared:                       0.436
                 coef    std err          t      P>|t|      [0.025      0.975]
const       -348.9463    122.900     -2.839      0.007    -596.054    -101.838
single        29.9911      4.920      6.095      0.000      20.098      39.884
Dep. Variable:                violent   R-squared:                       0.437
                 coef    std err          t      P>|t|      [0.025      0.975]
const       -344.7585    125.969     -2.737      0.009    -598.175     -91.342
poverty       -1.2906      6.573     -0.196      0.845     -14.514      11.933
single        30.5401      5.703      5.355      0.000      19.067      42.013
```

Leaving out single-parent households



Leaving out single-parent households



$$\underbrace{16}_{\beta'_{\text{pov}}} \approx \underbrace{-1.3}_{\beta_{\text{pov}}} + \underbrace{0.57 \times 30.5}_{\beta_{\text{pov,single}}\beta_{\text{single}} \approx 17.4}$$

`poverty` picks up the association that runs through single-parent households.

Who is the better hitter?

Batting average = hits / at bats, for two New York Yankees:

	1995	1996	both years
Derek Jeter	$12/48 = .250$	$183/582 = .314$	
David Justice	$104/411 = .253$	$45/140 = .321$	

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Question

Explain this in terms of omitted variable bias. What is the omitted variable?

Conditional means and the omitted year

Let $Y = 1$ for a hit and $Y = 0$ otherwise, $X_{\text{jeter}} = 1$ for Jeter (0 for Justice), and $X_{\text{year}} = 1$ for 1996 (0 for 1995). We consider a model

$$E[Y \mid X_{\text{jeter}}, X_{\text{year}}] = \beta_0 + \beta_{\text{jeter}} X_{\text{jeter}} + \beta_{\text{year}} X_{\text{year}}$$

where the table gives approximately

$$\begin{aligned}
\beta_0 &= E[Y \mid X_{\text{jeter}} = 0, X_{\text{year}} = 0] = 104/411 = .253, \\
\beta_0 + \beta_{\text{jeter}} &= E[Y \mid X_{\text{jeter}} = 1, X_{\text{year}} = 0] = 12/48 = .250, \\
\beta_0 + \beta_{\text{year}} &= E[Y \mid X_{\text{jeter}} = 0, X_{\text{year}} = 1] = 45/140 = .321, \\
\beta_0 + \beta_{\text{jeter}} + \beta_{\text{year}} &\approx E[Y \mid X_{\text{jeter}} = 1, X_{\text{year}} = 1] = 183/582 = .314.
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E[X_{\text{year}} \mid X_{\text{jeter}} = 1] &= 582/630 = .924, \\
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Both players hit better in 1996, and Jeter has more of his at bats that year. In the additive regression, this contributes to the coefficient when year is omitted:

$$\beta'_{\text{jeter}} = \beta_{\text{jeter}} + \beta_{\text{year}} \beta_{\text{jeter,year}}, \qquad \beta_{\text{jeter,year}} = .924 - .254 = .670.$$

Question

Recall the AR(1) process

$$Y_t = \rho Y_{t-1} + \epsilon_t.$$

Assume the process is stationary (all Y_t have the same marginal distribution). Suppose we regress Y_t on *two* previous values, Y_{t-1} and Y_{t-2}:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \epsilon_t'.$$

What should β_1 and β_2 be? Derive the regression coefficient when Y_{t-1} is omitted from the model using OVB. Then generalize this to obtain the regression coefficient between Y_{t-k} and Y_t.